

Statement of Work (SOW) M10 Booker Combat Vehicle Initial Operational Test (IOT)

NOTE: This SOW is a SAMPLE TEST for proposal action as specified at W51EW7-26-R-A004 and aligns with a SOW to be forwarded for Contractor action in accordance with the basic terms and conditions of the pending BTSS Follow-on contract. The Task Order Request for Proposal (TO RFP) shall be proposed as specified at Section L.

PERIOD OF PERFORMANCE: 120 Days

1. GENERAL TEST INFORMATION: Unless specifically changed by this SOW, the Contractor's performance shall be in accordance with the terms and conditions of the contract. If there is a conflict, the terms and conditions of the SOW shall take priority.

1.1. GENERAL TEST DATA

a. Test location/Dates:

Fort Stewart, GA (FSGA) (Planning Location)
Pilot Gunnery 2 Days
Record Gunnery 2 Days

Fort Bragg, NC (FBNC) (Planning Location)
Pilot Test Force-on-Force (FoF) 5 Days
Record Test (FoF) 14 Days

b. Other Dates:

New Equipment Training:

OPNET 1: 15 Days (Fort Bragg, NC) (Planning Location)
OPNET 2: 15 Days (Fort Bragg, NC) (Planning Location)

Field Level Maintainers New Equipment Training (FLMNET)
25 Days (Fort Bragg, NC) (Planning Location)

c. Classification: Unclassified

d. Technical Points of Contact:

Test Officer (TO), Maneuver Test Directorate (MTD).

Lead Operations Research System Analyst (ORSA), MTD.

Assistant ORSA, MTD

Assistant Test Officer (ATO), MTD.

Test Support Officer (TSO), G4.

2. CONTRACTING OFFICER REPRESENTATIVE:

- a. Contracting Officer's Representative (COR)
- b. Alternate Contracting Officer's Representative (ACOR)

3. TEST NARRATIVE INFORMATION:

3.1. System/Functional Description: The M10 BCV system provides early entry forces a mobile, protected, direct fire capability to apply immediate, lethal, long-range fires in the engagement of armored vehicles, hardened enemy fortifications, dismounted personnel and represents a long-term solution to the Infantry Brigade Combat Team capability. The M10 BCV capability is a new materiel solution, based on the Joint Requirements Oversight Counsel Capability Development Document approved.

Key BCV system characteristics include:

- Tracked vehicle with a minimum crew of four soldiers (commander, gunner, loader and driver).
- Two vehicles deployable by a single C-17 aircraft, in an operational, roll-on-roll-off (RORO) configuration.
- Crew protection from small arms, heavy machine guns, overhead artillery, underbelly mine, side improvised explosive device (IED), and kinetic energy (KE) threats.
- Utilizing ammunition available in the Department of Defense (DoD) inventory, the main gun applies precise, lethal, long-range fire, capable of neutralizing a bunker, conducting a wall breach and defeating light armor and second tier main battle tank equivalent armor while on the move, in day, night, and all-weather conditions.
- Sufficient platform speed and mobility to successfully keep pace with other vehicles in the formation.
- Communications system, which permits the simultaneous ability to transmit or receive both internal and external communications among all vehicle occupants.
- Sufficient size, weight, power, and cooling (SWaP-C) to facilitate integration of IBCT tactical network modernization technologies.
- Supported using the established logistics and maintenance structure for Army equipment, using the Army field level maintenance system with repair parts available through the existing supply system.
- 90% Operational Readiness (OR) rate.

3.2. Scope/Context: The M10 Booker Combat Vehicle (BCV) will fill a capability gap in the Infantry Brigade Combat Team (IBCT) identified in the Secretary of the Army's and

Chief of Staff (CoS) of the Army's Combat Vehicle Modernization Strategy. The BCV shall provide IBCTs a mobile, protected, direct fire capability to apply immediate, lethal, long-range fires in the engagement of armored vehicles, hardened enemy fortifications and dismounted personnel. The M10 BCV prevents the enemy from disrupting the momentum and tempo of the IBCT, ensuring freedom of movement and action. It also creates infantry breach points in urban, restricted, and open rolling terrain. The M10 BCV platoons support infantry battalions as they move, attack, defend, and perform other essential tasks to support the IBCT's mission. Infantry and BCV teams work together to bring the maximum combat power to bear on the enemy. The infantry locates and identifies targets for the BCV to engage, allowing infantry squads and platoons to maneuver along covered and concealed routes to assault enemy elements. Infantry squads provide protection for the BCV against attack by enemy infantry. The BCV provides heavy, continuous supporting fires against enemy strongpoints and positions. Infantry brigades lack an organic capability necessary to defeat enemy prepared positions and destroy enemy light armored vehicles ensuring freedom of maneuver and action for infantry in close contact with the enemy. The infantry brigade requires a protected, long range, precision direct fire capability to defeat enemy prepared positions, heavy machineguns, bunkers, and light armored vehicle threats during offensive operations or when conducting defensive operations against attacking enemy. The requirement for an M10 BCV capability is directly linked to the Army Warfighter Challenges (Conduct Joint Force Expeditionary Maneuver, Conduct Wide Area Security, Conduct Joint Combined Arms Maneuver, and Develop Capable Formations). Without this capability, infantry brigades confronting defending enemies in restrictive or urban terrain shall lose momentum, creating opportunities for the enemy and making static infantry forces vulnerable to enemy indirect fire and counterattacks. Because our adversaries shall operate in restrictive terrain and intermingle with civilian populations, indirect and aerial delivered fires shall prove inadequate to support infantry in close combat during offensive and defensive operations. The mission of M10 BCV is to provide precision, lethal, long range fires capability to the Infantry Brigade to quickly defeat enemy forces, destroy bunkers, walls, and light armored vehicles while maintaining momentum and freedom of action. Once tasks are organized, BCV platoons and sections move, attack, defend, and perform other specific tasks to support the brigade's mission. M10 BCV shall be employed based on conditions affecting METT-TC (I) as a company, platoon, or section. Normally, M10 BCV platoons shall be employed in support of Infantry Battalions and Companies. When METT-TC (I) conditions warrant, M10 BCV may also be employed dispersed in sections of two in direct support of infantry companies/platoons. In accomplishing its assigned missions, it employs firepower and maneuver, synchronizing its capabilities with those of other maneuver elements. The M10 BCV offers an array of capabilities on the modern battlefield. It can move rapidly in a variety of terrain conditions, negotiating soft ground, shallow trenches, small trees, and limited obstacles allowing the BCV to close with and destroy the enemy providing sustained, long range, precision lethal effects in all weather and light conditions. The M10 BCV does not change or alter the infantry brigade's Mission Essential Tasks (METL). This capability along with other future capabilities are aimed at enhancing the infantry brigade's ability to perform its METL in a complex environment.

4. SPECIFIC TECHNICAL REQUIREMENTS: The Contractor's response shall describe the technical approach and methodology with associated labor and reimbursable for each of the below sub-paragraphs. Cost shall be separately identified by each month.

4.1 Data Management:

Gunnery and 1st Excursion (Wall Breach and Bunker Neutralization)

a. Period of Support: 120 Days

b. Requirements:

(1) Attend and take part in meetings, conferences, technical exchange meetings, data collection requirement meetings, OTRR 3, and other test team meetings.

(2) Review documents provided by the government to resource requirements, such as the Operational Test Plan (OTP), the Pattern of Analysis (PA), and the Detailed Test Plan (DTP).

(3) Provide additional details with a data management plan that is in compliance with the data management methodology described in the draft OTP for the M10 Booker IOT&E. Which includes all processes and procedures by which digital and manually collected data is managed from collection through archival.

(4) Provide two separate databases: One for collecting and processing HSI and Performance type data and one for processing RAM data. The Enterprise Web Survey Suite (EWSS) shall be used for processing performance and HSI data. The ATOM database shall be used for the collection and processing of RAM type data.

(5) Provide input as appropriate on the Performance and HSI databases, which shall be used to collect, quality review, and authenticate Effectiveness and HSI data. Recommendations for modifications.

(6) Provide a data collector training plan and provide the training to data collectors on data collection and quality review procedures. The data collector-training plan shall be completed prior to the start of New Equipment Training (NET).

(7) Provide OPNET Gunnery populated databases and data dictionaries in support of all database delivery incrementally, daily during test and final delivery not later than 5 days after record test ends.

(8) Provide Force On Force populated databases and data dictionaries in support of all database delivery incrementally, daily during test and final delivery not later than 5 days after record test ends.

(9) Provide support for rehearsal of concept drills on data management, data collection, and data entry processes prior to test execution with ramp-up exercises to validate training and perform demonstrations with practical exercises to prepare for data collection.

(10) Perform quality review check of data to ensure completeness and accuracy. Resolve data anomalies and incorporate findings.

(11) Provide daily status reports depicting the number of tasks or functions executed from which data is derived, quantities of data records received and authenticated, and any current data anomalies.

(12) Provide support to each Data Authentication Group (DAG), RAM Scoring Conference (RSC), MERT and Focus Group meetings. Support shall consist of recording details of data anomalies, performing investigation and resolution of data anomalies, and tracking Request for Information (RFI) items. Provide a timeline of events during the MERT and Focus Groups to signify changes in topics. Provide a process, which shall track RFIs and provide this information to the ORSA daily. RFIs shall be resolved NLT 24 hours after submission.

(13) Provide a DAG viewer-training plan and provide DAG viewer training to DAG members in accordance with the procedures defined in the DAG Standard Operating Procedures (SOP), found in Appendix B to the M10 Booker IOT OTP provided by the government. Ensure associated procedures are included in the Data Management Plan.

(14) DAG training shall include, but is not limited to, procedures for accessing, voting, submitting comments and RFI using the data authentication method. Training material shall be made available in a hard copy for those who may need to refer to training as they authenticate data.

(15) Provide data collection support to collect Performance, HSI, and RAM data. Each data collector shall collect daily status timelines and failures as needed for maintenance actions. Provide RAM coverage on SUTs and the spare vehicles from the onset of baseline through the completion of Record Test at Fort Stewart and Fort Liberty.

(16) Data collectors shall be required to dismount and travel to the vehicle, once it stops, on foot while in support of the tactical mission based on daily concept of operations (CONOPS). During night movement, data collectors shall be required to conform to military night discipline operations.

(17) Perform RAM baselines and end of test final inventory for each SUT and the associated Supply Support Packages (SSP) prior to, and at the end of Gunnery, Pilot and Record Test.

(18) Perform quality checks for the completeness of all data collected and for the validation of data in accordance with the approved test design. Provide quality control and run edit checks on all data.

(19) Provide daily reports to ORSA in accordance with Chapter 2 of the M10 Booker IOT OTP. Daily reports include but are not limited to total system operating hours RAM, number of open and closed TIRs, the quantity of data from each harvest that is in reduction and the quantity that has been authenticated.

(20) Provide after action and lessons learned report containing comments and recommendations on procedures, and events pertaining to data management, data collection, and report preparation. Provide as input to the Test Team's After-Action Report (AAR).

(21) Archive test data for delivery to OTC Test Team.

(a) Scanned Raw Data Forms (if applicable): Verify the scanned copy is readable and an exact copy of the original. Save the scanned forms and make them retrievable as if they were stored in paper form (manila folders). Once the verification process is completed and with approval from the Test ORSA, the paper forms shall be shredded.

(b) Electronic/Digital Records: If the document is received electronically or in digital format, then keep it that way. No need to print the document for archiving - it can be archived in electronic format.

(c) When archiving data, keep test-planning files separate from raw data files; for example, use separate disks.

(d) Back-up test data files daily on an external storage device.

NOTE: Data collection will start with the Initial RAM Baseline for Gunnery. 6 SUTs (5 active and 1 spare) in support of gunnery. Gunnery; performed during day and night runs. RAM data shall be collected for the entire duration of the gunnery. HSI surveys and Focus Group data shall be collected during the execution of Tables V and VI. Data shall be collected periodically (Anticipate 12-16 hours per day) during gunnery. The Data Management (DM) Cell shall not perform 24-hour operations during gunnery but may require extended hours of support to meet data reduction and delivery deadlines established by test team.

(22) During the execution of the 1st Excursion (Wall Breach & Bunker Neutralization). Data shall be collected by Aberdeen Test Center (ATC) personnel. OTC contractor shall gather the operational data and administer any observation cards if required. 2 SUTs with crews will perform the excursion.

(23) Provide support for a daily data harvest during the execution of the gunnery. The collection of RAM data, photos supporting TIRs and Test observations shall be the

bulk of the data harvest. RAM starter cards shall be provided to all crew members in the SUT to support TIR data collection. Administration of gunnery surveys shall follow.

(24) Support for 1 Focus Group during the conclusion of the gunnery (Tables V and VI) shall be in the form of writing down a time and a subject of discussion or short narrative, while the event is being recorded. The time and short narrative shall be used to provide a time stamp and a subject of the interviews. The time record shall aid the AEC evaluator with faster look-up of Soldier responses.

(25) At the completion of the Gunnery and 1st Excursion (Wall Breach and Bunker Neutralization) process the gunnery data and prepare it for delivery.

(26) Update DC forms (HSI, Performance, RAM) and electronic tablets with the DC forms then prepare electronic tablets in preparation of the upcoming FoF.

(27) Review the pros and cons of the data form administration, collection and harvest conducted at Gunnery and see what can be improved.

(28) Provide training and practical exercises in the usage of electronic media for collecting RAM, Performance and HSI data in support of data collection of two data excursions during the execution of the FoF.

NOTE: Data excursions are separate events from the overall DC process and are conducted once, possibly twice, so the data collection must be right all the time. There is no make-up of excursion data.

(28) Provide data harvest rehearsals to support the DC training.

(29) Provide a methodology to track RAM vehicle miles, engine hours, and other metered vehicle data on the M10 Booker test/spare vehicles.

NOTE: This test is a RAM intensive test, among other data requirements, that shall require the collection of accurate and completed RAM data starting at the Gunnery phase followed by the pilot test phase and will continue non-interrupted thru end-to-end data run and into the record test phase. Limited M10 Booker vehicle movement during the end-to-end data run but full RAM shall be collected during that time. Frequency of data collection during that time shall be once every 24 hrs. The tracking on an Excel spreadsheet of each of the M10 Booker vehicles plus spares the RAM miles, engine hours and other metered vehicle data is crucial and done daily. The mileage shall be used for adjusting the daily missions based on the miles accumulated.

(30) Thirteen Systems Under Test (SUT) shall be in used the FoF. 10 SUTs will be in play, 2 will be spares and 1 will provide support to the Cyber. RAM data shall be collected on all 13 SUTs. The SUT supporting Cyber shall require Data Collector support and will mostly be stationary during the execution of the FoF. The two spares will become active in the event that any of the 10 SUTs suffers a maintenance failure that will prevent it from returning to the unit within 24 hrs.

(31) Anticipate up to 24-hour daily coverage for RAM data collection at the field maintenance location. The location will be manned by the unit maintainers (Soldiers) and Field Service Representatives (FSR) as required. RAM/Maintenance data shall be collected on any SUT that arrives at this location. Photos shall be taken at this location in support of RAM data collection. Maintainers shall rest sometime during the night which RAM coverage can be suspended until the next start. The actual time frame will not be known until the unit provides the hours of maintenance support they plan to work.

(32) Anticipate 24-hour coverage for RAM data collectors supporting the 12 SUTs in play during FoF mission execution. (The SUTs break down: 10 SUTs in play plus 2 Spares) The Data Management (DM) Cell shall not perform 24-hour operations during the FoF but may require extended hours of support to meet data reduction and delivery deadlines established by test team.

(33) Provide support for a daily data harvest during the execution of the pilot and record test of the FoF with the collection of RAM data, photos, and administration of the performance and HSI surveys by electronic means.

(34) Roll On/Roll Off (RO/RO) Fly away excursion; Two SUTs with crews, will support in the execution of this excursion. This will be a timed event with two SUTs being loaded onto a C17 and will fly off from Pope Army Airfield and will download at Holland Landing Zone. RAM data, and performance data shall be collected during this excursion. Expect photos to be taken of any issue that may arise.

(35) Towing Excursion: During the FoF there shall be a towing of two SUTs and RAM and other pertinent data collected. If towing does not occur under normal conditions, then two separate towing events will be executed by Test Operations during the FoF to exercise the towing capability of the M10 booker.

(36) All Contractor personnel must have and maintain a CAC card.

NOTE: Anticipate 24-hour coverage for only RAM data collectors supporting

(37) Data collection personnel shall ride in government vehicles with a government person driving, two per vehicle to cover one-M10 Booker (2 x M10 Booker vehicles). Be prepared to operate at night maintaining light discipline in order to maintain the test unit's field tactical posture.

c. Milestones/Deliverables:

Event/Requirement
Pre-Brief EDR
OPNET 1
Event Design Review (EDR)
FLMNET 1

OPNET 2
FLMNET 2
OTRR 2
OPNET Gunnery
Training Data Collector (Fort Hood, TX)
Data Collection 1 st Rehearsal (Gunnery)
Test Team Deploys to Fort Stewart, GA
DAG Training, Fort Stewart, GA
IOT&E Gunnery
Gunnery ROC Drill
Data Collection Walk thru (Fort Stewart, GA)
Pilot Test (Gunnery)
Excursion 1 Wall Breach and Bunker Neutralization, Aberdeen Proving Ground, MD
OTRR 3A
Record Test Gunnery
Record Test (Gunnery) Data closeout/RAM Scoring Conference/DAG
Redeploy Test Team to Fort Hood, TX
Deploy Test Team to Fort Bragg, NC
Training Data Collector, Fort Bragg, NC
Data Collector Rehearsal, Data team only
Data Collector Rehearsal and Walk thru with Test OPS
DAG Training, Fort Bragg, NC
RAM Initial Baseline (FoF)
Pilot Test (FoF)
End to End Data Run (FoF)
C17 Roll On/Roll Off Fly away Excursion II
Record Test
Record Test Data closeout/RFI/RSC/DAG
Redeploy Test Team to Fort Hood, TX
Provide populated database and data dictionaries

4.2 LOGISTICS:

- a. Period of Support: 120 Days
- b. Requirements:

(1) Prepare for issue in-accordance-with (IAW) BTSS Property Management, Standing Operating Procedures (SOP) to POC the below listed Government Property (GP.) DA form 1687 shall be prepared by the G4, TSO, signed by the Test Officer, which shall authorize select test team personnel to request/receive GP.

(2) Provide Night Vision Device (NVD) training to Military and Government civilian personnel. NVDs shall be used to support Dismounted Operations during Pilot and Record test.

Item Description	Qty.
GPS Handheld w/Extra Battery	25
Night Vision Goggles - PVS 14	25
Radio Set - AN/VRC-92 - SINCGARS	2
Antenna Repeater	1
Data Collector Kit	60
Eye Protection	20
Helmet, Ground Troop, Kevlar w/Accessories	20
NVG Helmet Mount	25
Shredder, Unclassified	4
Base Station w/Pwr/Ant, Commercial Handheld Radio	1
Battery, Commercial Handheld Radio	75
Radio Commercial Handheld w/Battery	75
Charger, Commercial Handheld Radio, 6 Unit	10
Charger, Commercial Handheld Radio, 1 Unit	15
Refrigerator, 14 Cu Ft	4
Bull Horn	2
Chair, Folding	100
Cooler, Igloo 5 Gallon	50
Ice Chest, Igloo 72 Quart	50
Dry Erase Board, 36" x 48"	4
Extension Cord, 50 ft	8
Extension Cord, 100 ft	4
Extension Cord, 200 ft	2
Surge Protector	75
Table, Folding 5 ft	50

4.3. OPERATIONS:

a. Period of support: 120 Days

b. Requirements:

NOTE: Contractor personnel assigned this task shall work with both Government civilian employees as well as military personnel from the test team. A government representative shall be on duty during the entire test duration. All contractor personnel must have and maintain a CAC card.

(1) Provide operations cell support during setup, Pilot test, Record test, and tear down operations. Test operations cell shall be operational 24 hours a day from the onset of Pilot through the completion of Record tests.

(2) Maintain an electronic events form (similar to DA Form 1594) of daily test events. At the end of each day, forward the form to data management for entry into the database.

(3) Monitor radio traffic for Land Mobile Radio communications and document activities/events on the electronic events form.

(4) Update operations charts to include equipment status, personnel status, and daily weather charts.